

1. DATA PREPARATION STRATEGY

The quality of predictions depends directly on the rigor of the initial cleaning. Our pipeline follows a logic of maximum information retention while eliminating biases.

1.1 Filtering and Variable Removal

Before any modeling, we purified the dataset to ensure learning integrity:

- Noise elimination: Unique identifiers (`bcr_patient_barcode`) and administrative dates were removed as they have no biological causality with survival.
- Redundancy reduction: Variables such as `disease_type` or `primary_site`, being constant for all patients (only one type of cancer studied), provide no explanatory variance.
- Data leakage prevention: Critical deletion of the variable `follow_ups_disease_response`. This variable, strongly correlated with the target, contains information on future treatment response unavailable at the time of diagnosis (T_0). Including it would artificially skew the results.

1.2 Missing Value Treatment (NA)

Little's test ($p < 0.001$) rejected the MCAR (Missing Completely At Random) hypothesis, justifying a thoughtful imputation strategy rather than blind deletion:

- Semantic imputation: For `prior_treatment`, we considered that the absence of data signified an absence of treatment ("No").
- Explicit uncertainty: For complex categorical variables like `primary_diagnosis`, we created an "Unknown" class to avoid introducing bias through an arbitrary mode.
- Statistical approach: The median was used for continuous variables to minimize the impact of outliers.

1.3 Variable Preparation and Selection

- Null variance (`nearZeroVar`): The algorithm automatically excluded quasi-constant demographic variables (e.g., `gender`, `ethnicity`), confirming the homogeneity of the studied cohort.
- Transformation: The complex variable `sites_of_involvement` was split via One-Hot Encoding to allow the model to treat each metastatic site independently.

2. MODELING AND RESULTS

The modeling phase follows a progressive structure. We first validate the importance of data imputation using clinical baselines. Then, we explore different variable selection strategies (Lasso, Fisher, Limma) to manage the high dimensionality of genomic data.

2.1 Baseline: The Impact of Imputation

We compared a standard GLM trained on raw data (deleting missing rows) versus the imputed dataset.

A. Raw Clinical Data (Complete Case)

Prediction	Ref: Alive (0)	Ref: Dead (1)
Alive (0)	118	8
Dead (1)	68	22

Accuracy: 64.8% / Sensitivity: 73.3% / Precision: 24.4%

B. Imputed Clinical Data

Prediction	Ref: Alive (0)	Ref: Dead (1)
Alive (0)	191	22
Dead (1)	13	18

Accuracy: 85.7% / AUC: 0.718 / Sensitivity: 45.0% / Precision: 58.1%

Conclusion on Strategy: The imputed model drastically outperforms the raw model in terms of stability (Accuracy +20.9%). While the raw model appeared more sensitive (73.3%), it suffered from excessive noise, predicting "Dead" too often (Precision 24.4).

C. Baseline: GeneX Only (Naive GLM) We attempted a naive GLM on the full imputed genomic dataset without any prior variable selection.

Prediction	Ref: Alive (0)	Ref: Dead (1)
Alive (0)	99	23
Dead (1)	106	17

Accuracy: 47.4% / AUC: 0.546 / Sensitivity: 42.5% / Precision: 13.8%

Analysis: The results are catastrophic. With an AUC of 0.546, the model performs barely better than random chance, overwhelmed by noise (106 False Positives). This failure leads to a dual conclusion:

- 1. Dimensionality Reduction is mandatory:** We must apply screening (Lasso, Fisher, Limma) to filter out noise.
- 2. Hybridization is necessary:** Since genomic data alone lacks the predictive stability of the clinical baseline (Section B), we must **concatenate the selected genes with the cleaned clinical data** to combine biological precision with clinical robustness.

2.2 Screening via Lasso Regularization

Lasso (ℓ_1 penalty) performs embedded feature selection, shrinking non-predictive coefficients to zero. **A. Lasso on GeneX Only (glmnet)**

Prediction	Ref: Alive (0)	Ref: Dead (1)
Alive (0)	135	15
Dead (1)	70	25

Accuracy: 65.3% | AUC: 0.682 | Sensitivity: 62.5% | Precision: 26.3%

Analysis: Lasso successfully extracts a "genetic signature," raising accuracy from 47.4% (Naive GLM) to 65.3%. While it recovers signal from the noise, the precision remains low (26.3%), indicating that genomic data alone generates many False Positives compared to clinical data.

B. Lasso on Combined Data (Clinical + GeneX)

Prediction	Ref: Alive (0)	Ref: Dead (1)
Alive (0)	193	23
Dead (1)	11	17

Accuracy: 86.1% | Sensitivity: 42.5% | Precision: 60.7%

Analysis: Merging clinical context with Lasso-selected genes drastically improves reliability. The Accuracy jumps to 86.1%. Most importantly, the **Precision more than doubles** compared to the GeneX-only model (60.7% vs 26.3%). The model successfully filtered out the "genomic noise," reducing False Positives to only 11. Although Sensitivity (42.5%) is lower than the pure genomic Lasso, the predictions made are far more trustworthy.

2.3 Screening via Hybrid Stepwise (AIC)

Directly applying Stepwise selection on thousands of genes is computationally infeasible and prone to massive overfitting. We adopted a hybrid strategy: first pre-selecting the top 200 genes via univariate T-tests, followed by a bidirectional Stepwise selection (minimizing AIC) to refine the model.

Prediction	Ref: Alive (0)	Ref: Dead (1)
Alive (0)	189	26
Dead (1)	15	12

Accuracy: 83.0% | AUC: 0.684 | Sensitivity: 31.6% | Precision: 44.4%

Analysis: While the Accuracy (83.0%) appears high, it is driven by the model's tendency to predict the majority class ("Alive"). The Specificity is excellent (92.6%), but the Sensitivity is poor (31.6%). The model misses the vast majority of deaths (26 False Negatives vs 12 True Positives). This confirms the limitations of Stepwise selection in genomics: the algorithm handles multicollinearity poorly, arbitrarily discarding correlated genes that could have improved sensitivity.

3. SYNTHESIS AND CONCLUSION

This hierarchy of models allows us to determine the optimal method for integrating high-dimensional genomic data.

Strategy	Method	Acc.	Sens.
1. Baseline	Clinical (Raw)	64.8%	73.3%
1. Baseline	Clinical (Imputed)	85.7%	45.0%
2. Lasso	GeneX Only	65.3%	62.5%
2. Lasso	Combo (Clin+Gene)	86.1%	42.5%
3. Fisher	Filter Screening	74.8%	36.8%
4. Limma	Diff. Expression	73.1%	44.7%

Final Conclusion: The Lasso on Combined Data emerges as the optimal model, maximizing reliability (Accuracy 86.1%, Precision 60.7%). However, the ceiling in performance—particularly the low Sensitivity (42.5%)—is likely

2.4 Screening via Fisher Test

We applied a univariate filter using Fisher's exact test (or F-score) to rank genes based on their individual correlation with survival, independent of the model.

Model Performance (Fisher Top-K):

Prediction	Ref: Alive (0)	Ref: Dead (1)
Alive (0)	167	24
Dead (1)	37	14

Accuracy: 74.8% | Sensitivity: 36.8% | Precision: 27.5%

Analysis: While Fisher screening yields a higher Accuracy (74.8%) than Lasso GeneX (65.3%), it fails at the critical task: Sensitivity drops drastically to 36.8%. The model becomes conservative, predicting "Alive" for most patients. This highlights the limitation of univariate filtering: by analyzing genes in isolation, we lose the predictive power of gene interactions that Lasso was able to capture.

2.5 Screening via Limma

Note: Differential Expression Analysis.

Limma (Linear Models for Microarray Data) uses empirical Bayes moderation to assess differential expression. We selected the top 200 genes with the most significant Fold Change between "Alive" and "Dead" patients.

Model Performance (Limma Selection):

Prediction	Ref: Alive (0)	Ref: Dead (1)
Alive (0)	160	21
Dead (1)	44	17

Accuracy: 73.1% | AUC: 0.629 | Sensitivity: 44.7% | Precision: 27.9%

Analysis: Limma offers a slight improvement over Fisher in terms of Sensitivity (44.7% vs 36.8%), likely due to the empirical Bayes moderation which stabilizes variance estimation. However, it still underperforms compared to Lasso (Sensitivity 62.5%). This reinforces the finding that univariate selection methods (Fisher, Limma) are less effective at capturing complex multi-gene patterns than multivariate embedded methods like Lasso.

attributable to our clinical feature engineering. The conservative strategy of imputing missing values with generic classes (e.g., "Unknown") and grouping rare categories likely "smoothed out" specific high-risk signals, preventing the model from detecting the most critical profiles.